Issue Tracking Management System

Database Module Architecture Document

This document presents the database module architecture of the Enterprise Issue Tracking Management System. The database has been designed using relational database principles, normalization techniques, and modular architecture to ensure scalability, maintainability, security, and performance.

The architecture separates the system into independent functional modules. Each module is responsible for a specific business function while maintaining strong relationships with other modules through well-defined foreign key constraints. This modular design improves system organization, simplifies maintenance, and supports future expansion without requiring significant structural changes.

Module 1: System Configuration (Master Data)

The System Configuration module provides centralized management of all reference (lookup) data used throughout the application. Instead of embedding fixed values within the application code, business reference data is maintained in dedicated tables, ensuring consistency, flexibility, and ease of administration.

Responsibilities

- ✓ Configure user roles
- ✓ Configure account statuses
- ✓ Configure issue categories
- ✓ Configure issue priorities
- ✓ Configure issue statuses
- ✓ Configure approval statuses
- ✓ Configure assignment statuses
- ✓ Configure assignment types
- ✓ Configure notification types
- ✓ Configure attachment types
- ✓ Configure audit actions
- ✓ Configure escalation statuses

Database Tables

- ✓ roles
- ✓ account_statuses
- ✓ issue_categories
- ✓ priorities
- ✓ issue_statuses
- ✓ approval_statuses
- ✓ assignment_statuses
- ✓ assignment_types
- ✓ notification_types
- ✓ attachment_types
- ✓ audit_actions
- ✓ escalation_statuses

Business Value

This module standardizes business data across the entire application, improves data integrity, eliminates redundancy, and supports future business rule modifications without changing application logic.

Module 2: Organization Management

The Organization Management module represents the organizational structure of the company. It enables the system to organize departments under organizations and provides the foundation for user management and issue ownership.

Responsibilities

- ✓ Register organizations
- ✓ Maintain organizational information
- ✓ Create and manage departments
- ✓ Support multi-organization environments
- ✓ Organize business units

Database Tables

- ✓ organizations

- ✓ departments

Relationship

One organization may contain multiple departments, while each department belongs to one organization.

Business Value

Provides a scalable organizational structure suitable for companies with multiple departments or multiple business entities.

Module 3: User & Access Management

This module manages all system users and controls authentication and authorization. It ensures that users access only the functions permitted by their assigned roles.

Responsibilities

- ✓ Register users
- ✓ Manage user accounts
- ✓ Assign departments
- ✓ Assign roles
- ✓ Manage account status
- ✓ Authenticate users
- ✓ Authorize system access
- ✓ Maintain employee profiles

Database Tables

- ✓ users
- ✓ roles
- ✓ account_statuses
- ✓ departments

Relationship

Each user belongs to one department, has one role, and maintains one account status.

Business Value

Provides secure access control, centralized user administration, and role-based authorization throughout the application.

Module 4: Issue Management

Issue Management is the core business module responsible for recording, tracking, updating, and resolving reported issues throughout their lifecycle.

Responsibilities

- ✓ Register new issues
- ✓ Maintain issue information
- ✓ Categorize issues
- ✓ Assign priorities
- ✓ Track issue status
- ✓ Record issue requester
- ✓ Monitor issue lifecycle
- ✓ Close resolved issues

Database Tables

- ✓ issues
- ✓ issue_categories
- ✓ priorities
- ✓ issue_statuses
- ✓ users
- ✓ departments

Relationship

Each issue belongs to one requester, one department, one category, one priority, and one current status.

Business Value

Provides complete visibility into issue processing while ensuring every issue can be monitored from creation through resolution.

Module 5: Assignment Management

This module manages technician assignments and controls how work is distributed throughout the organization.

Responsibilities

- ✓ Manual assignment
- ✓ Automatic assignment (handled by backend business logic)
- ✓ Technician reassignment
- ✓ Assignment status management
- ✓ Assignment history
- ✓ Technician workload tracking

Database Tables

- ✓ assignments
- ✓ assignment_types
- ✓ assignment_statuses
- ✓ users
- ✓ issues

Relationship

One issue may have multiple assignment records during its lifecycle, while one technician may manage multiple issues.

Business Value

Ensures efficient workload distribution and complete traceability of assignment activities.

Module 6: Approval Workflow Management

The Approval Workflow module controls managerial approval processes required before issues proceed to the next business stage.

Responsibilities

- ✓ Submit issues for approval
- ✓ Record approval decisions
- ✓ Support multi-level approval
- ✓ Record approvers

- ✓ Maintain approval history
- ✓ Track approval status

Database Tables

- ✓ approvals
- ✓ approval_statuses
- ✓ issues
- ✓ users

Relationship

Each issue may require one or more approval records depending on organizational workflow requirements.

Business Value

Improves governance, accountability, and compliance by ensuring appropriate authorization before critical business actions are completed.

Module 7: SLA & Escalation Management

This module ensures that all issues are handled within agreed Service Level Agreements (SLAs) and automatically escalated when deadlines are exceeded.

Responsibilities

- ✓ Define SLA rules
- ✓ Monitor response deadlines
- ✓ Monitor resolution deadlines
- ✓ Track escalation status
- ✓ Record escalation history
- ✓ Support SLA compliance monitoring

Database Tables

- ✓ sla
- ✓ escalations
- ✓ escalation_statuses
- ✓ priorities

- ✓ issues

Relationship

SLA rules are associated with issue priorities, while escalation records are associated with individual issues.

Business Value

Improves service quality by monitoring performance targets and ensuring timely issue resolution.

Module 8: Communication Management

This module maintains all communication associated with issue resolution, including comments, discussions, and supporting documents.

Responsibilities

- ✓ Record issue comments
- ✓ Upload supporting documents
- ✓ Upload screenshots
- ✓ Maintain discussion history
- ✓ Store communication evidence

Database Tables

- ✓ comments
- ✓ attachments
- ✓ attachment_types
- ✓ issues
- ✓ users

Relationship

Each issue may contain multiple comments and multiple supporting attachments.

Business Value

Provides centralized communication history and preserves evidence related to issue investigation and resolution.

Module 9: Notification Management

Objective

The Notification Management module informs users of important business events throughout the issue lifecycle.

Responsibilities

- ✓ Assignment notifications
- ✓ Approval notifications
- ✓ Issue status notifications
- ✓ Escalation notifications
- ✓ Reminder notifications
- ✓ Notification history

Database Tables

- ✓ notifications
- ✓ notification_types
- ✓ users
- ✓ issues

Relationship

Notifications are generated for users based on events associated with issues.

Business Value

Improves communication, increases user awareness, and supports timely completion of assigned responsibilities.

Module 10: Audit & History Management

This module records every significant system activity to provide complete traceability, accountability, and compliance.

Responsibilities

- ✓ Record user activities
- ✓ Maintain audit logs
- ✓ Track issue history
- ✓ Record status changes
- ✓ Record assignment changes

- ✓ Record approval activities
- ✓ Support compliance auditing

Database Tables

- ✓ audit_logs
- ✓ audit_actions
- ✓ issue_history
- ✓ users
- ✓ issues

Relationship

Audit logs record user activities, while issue history records lifecycle changes associated with each issue.

Business Value

Provides accountability, operational transparency, regulatory compliance, and historical analysis capabilities.

Module 11: Reporting & Analytics

This module transforms operational data into meaningful business information through dashboards, statistics, and management reports.

Responsibilities

- ✓ Operational dashboards
- ✓ Issue statistics
- ✓ Technician performance reports
- ✓ Department performance reports
- ✓ SLA compliance reports
- ✓ Resolution time analysis
- ✓ Approval reports
- ✓ Executive management reports

Database Tables

This module does not require dedicated database tables. It retrieves and aggregates information from operational tables including:

- ✓ issues
- ✓ assignments
- ✓ approvals
- ✓ escalations
- ✓ notifications
- ✓ audit_logs
- ✓ issue_history
- ✓ users
- ✓ departments
- ✓ organizations

Business Value

Supports management decision-making through accurate reporting, performance monitoring, and business intelligence.

Overall Database Architecture

The Enterprise Issue Tracking Management System is organized into eleven integrated modules, each responsible for a distinct business function. Together, these modules provide a comprehensive, scalable, and maintainable database architecture suitable for enterprise applications.

The database follows Third Normal Form (3NF), employs lookup tables for configurable business values, enforces referential integrity through foreign keys, and supports auditing, notifications, approval workflows, SLA management, and reporting. The design is intended to support future growth, simplify maintenance, and integrate effectively with the Node.js/Express backend and React frontend.

This modular architecture aligns with modern enterprise software engineering practices by separating concerns, reducing redundancy, improving maintainability, and providing a strong foundation for secure, high-performance application development.